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1.5 Quiz: Arithmetic Series

1. [Maximum mark: 7]

In an arithmetic sequence, $u_4 = 17$ and $u_9 = 37$.

(a) Find the value of the common difference d and the value of the first term u_1 . [3]

(b) Hence write down an expression for u_n . [1]

(c) Determine whether 200 is a term of this sequence. Justify your answer. [3]

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2. [Maximum mark: 6]

A theatre has rows of seats. The first row has 18 seats. Each row after the first has 3 more seats than the row in front of it. The number of seats in row n is u_n .

- Find an expression for u_n in terms of n . [2]
- Find the number of seats in row 12. [1]
- Find the number of the first row that has more than 65 seats. [3]

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3. [Maximum mark: 5]

Consider the arithmetic series

$$5 + 8 + 11 + \cdots + 62.$$

(a) Write down the value of the common difference d . [1]

(b) Find the number of terms in the series. [2]

(c) Find the sum of the series. [2]

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4. [Maximum mark: 7]

Luca saves money every week. In week 1 he saves \$5. Each week after that, he saves \$2 more than he saved the week before.

(a) Find the amount Luca saves in week 12. [2]

(b) Find the total amount Luca has saved by the end of week 12. [2]

(c) Find the first week when Luca's total savings are more than \$200. [3]

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5. [Maximum mark: 5]

The first three terms of an arithmetic sequence are

$$u_1 = 2k, \quad u_2 = 3k + 1, \quad u_3 = 5k - 2.$$

(a) Show that $k = 4$. [2]

(b) Write down the first three terms of the sequence. [1]

(c) Find the sum of the first 10 terms of the sequence. [2]

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6. [Maximum mark: 5]

The first term of an arithmetic sequence is $u_1 = 4$. The sum of the first 10 terms of the sequence is $S_{10} = 175$.

(a) Find the value of the common difference d . [3]

(b) Find S_{20} , the sum of the first 20 terms. [2]